APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – III

Incompressible Viscous Flows and Compressible Flows

Syllabus (as given)

- Formulations of incompressible viscous flows by finite difference methods, pressure correction methods, vortex methods.
- Treatment of compressible flows: potential equation, Euler equations, Navier–Stokes system of equations, flow-field-dependent variation methods, boundary conditions, example problems.

1. Incompressible Viscous Flow — Governing Equations

For constant density ρ and viscosity μ (kinematic viscosity $\nu = \mu/\rho$):

$$\nabla \cdot \mathcal{V} = 0 \qquad \text{(continuity)}$$

$$\partial \mathcal{V}/\partial t + (\mathcal{V}\cdot\nabla)\mathcal{V} = -(1/\rho)\nabla p + \nu\nabla^2\mathcal{V} + f \qquad \text{(momentum)}$$

In 2-D Cartesian form:

$$u_x + v_y = 0$$

$$u_t + u\,u_x + v\,u_y = -p_x/\rho + \nu(u_{xx} + u_{yy})$$

$$v_t + u\,v_x + v\,v_y = -p_y/\rho + \nu(v_{xx} + v_{yy})$$

Difficulties of the incompressible equations

- There is no evolution equation for pressure — p acts as a Lagrange multiplier that enforces $\nabla\cdot\mathcal{V} = 0$, so continuity becomes a constraint, not a time-marching equation.
- Velocity–pressure coupling: u, v, p must be solved together; naive central differencing on a collocated grid leads to checkerboard pressure (odd–even decoupling).
- Non-linear convective terms ($u\,u_x$) and large Reynolds numbers $Re = UL/\nu$ need careful differencing (central vs upwind, cell Reynolds number).
- Two main strategies: (i) eliminate pressure — vorticity–stream function; (ii) keep primitive variables (u, v, p) — MAC, projection, SIMPLE (pressure correction).

2. Vorticity–Stream Function Formulation (2-D)

Define stream function ψ and vorticity ω:

$$u = \partial\psi/\partial y, \quad v = -\partial\psi/\partial x \qquad \text{(continuity automatically satisfied)}$$

$$\omega = \partial v/\partial x - \partial u/\partial y = -\nabla^2\psi$$

Take the curl of the momentum equations (pressure disappears):

$$\partial\omega/\partial t + u\,\partial\omega/\partial x + v\,\partial\omega/\partial y = \nu\,(\partial^2\omega/\partial x^2 + \partial^2\omega/\partial y^2) \qquad \text{(vorticity transport)}$$

$$\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = -\omega \qquad \text{(Poisson equation for } \psi\text{)}$$

Finite-difference discretisation

$$\omega_{ij}^{n+1} = \omega_{ij}^{n} - \Delta t[u\,(\omega_{i+1,j} - \omega_{i-1,j})/2\Delta x + v\,(\omega_{i,j+1} - \omega_{i,j-1})/2\Delta y] + \nu\Delta t[\delta_x^2\omega/\Delta x^2 + \delta_y^2\omega/\Delta y^2]$$

$$\psi_{i+1,j} + \psi_{i-1,j} + \psi_{i,j+1} + \psi_{i,j-1} - 4\psi_{i,j} = -h^2\,\omega_{ij} \qquad (\Delta x = \Delta y = h)$$

$$u_{ij} = (\psi_{i,j+1} - \psi_{i,j-1})/2\Delta y\,, \qquad v_{ij} = -(\psi_{i+1,j} - \psi_{i-1,j})/2\Delta x$$



Wall vorticity (Thom's formula)

At a stationary wall ψ = const (no flux) and $u = v = 0$. Expand ψ near the wall (wall at $j = 1$):

$$\omega_{wall} = -2(\psi_{wall+1} - \psi_{wall})/\Delta n^2 \qquad \text{(stationary wall)}$$

Moving lid with speed U: $\omega_w = -2(\psi_1 - \psi_w)/\Delta n^2 - 2U/\Delta n$

- Δn = distance between the wall and first interior node. Second-order accurate version uses three points.

Pros and cons of ψ–ω

- Pressure eliminated, continuity satisfied exactly, only 2 unknowns, good for 2-D cavity/channel flows.
- Wall vorticity BC is the weak point; not directly extendable to 3-D (vector potential needed); multiply-connected domains need extra conditions; pressure must be recovered separately by solving a Poisson equation.

Convective-term differencing in Navier–Stokes solvers

Scheme	Formula (u>0)	Remark
Central	$(u_{i+1} - u_{i-1})/2\Delta x$	2nd order; needs $Re_\Delta \le 2$
1st-order upwind	$(u_i - u_{i-1})/\Delta x$	stable, diffusive
2nd-order upwind	$(3u_i - 4u_{i-1} + u_{i-2})/2\Delta x$	less diffusive, wiggles
QUICK	$(3u_{i+1} + 3u_i - 7u_{i-1} + u_{i-2})/8\Delta x$	3rd order
Hybrid	central if $\|Pe\| < 2$, else upwind	robust

- Quasi-linear momentum equation: coefficient of u is itself u — treat by lagging (Picard) or Newton linearisation, with under-relaxation.
- Explicit stability limits for 2-D: $\Delta t \le \min(\Delta x^2/4\nu\ ,\ 4\nu/(u^2+v^2))$ for FTCS-type, and CFL $(|u|/\Delta x + |v|/\Delta y)\Delta t \le 1$ for upwind.

Worked example: wall vorticity

Thom formula — numbers

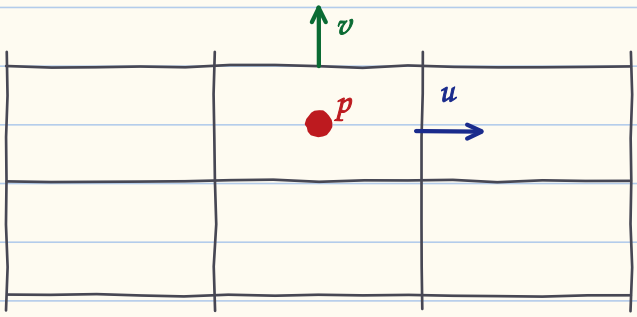
Stationary wall at $y = 0$, $\psi_{wall} = 0$, first interior node at $\Delta y = 0.1$ where $\psi_1 = 0.0004$.

$\omega_{wall} = -2(\psi_1 - \psi_{wall})/\Delta y^2 = -2(0.0004)/0.01 = -0.08$.

Moving lid $U = 1$ at the top, $\psi_{N-1} = -0.002$ (below the lid), $\Delta y = 0.1$: $\omega = -2(-0.002)/0.01 - 2(1)/0.1 = +0.4 - 20 = -19.6$.

3. Primitive-variable Formulations

3.1 Staggered grid (Harlow–Welch)



p at cell centre

u at east/west faces

v at north/south faces

→ avoids checkerboard p

- Pressure at cell centres, u at vertical faces, v at horizontal faces -> pressure difference drives the velocity located between the two pressure nodes; discrete continuity involves compact differences

$(u_{i+1/2} - u_{i-1/2})/\Delta x + (v_{j+1/2} - v_{j-1/2})/\Delta y$.

- Eliminates checkerboard oscillation; widely used in finite-difference and finite-volume codes (alternative: collocated grid + Rhie–Chow interpolation).

3.2 Pressure Poisson equation

Take divergence of momentum equation and use $\nabla \cdot \mathcal{V} = 0$:

$$\nabla^2 p = -\rho \left[(\partial u/\partial x)^2 + 2\,(\partial u/\partial y)(\partial v/\partial x) + (\partial v/\partial y)^2 \right] \quad (+\,\partial D/\partial t \text{ term in MAC})$$

- With Neumann conditions on walls $\partial p/\partial n = \mu\, \nabla^2 \mathcal{V} \cdot n$. Solved by SOR / multigrid at every step.

3.3 MAC (Marker-And-Cell) method

1. Given $\mathcal{V}^n$ (with $\nabla \cdot \mathcal{V}^n = 0$), compute pressure p^n from the Poisson equation.
2. Advance momentum explicitly: $\mathcal{V}^{n+1} = \mathcal{V}^n + \Delta t[-(\mathcal{V} \cdot \nabla)\mathcal{V} - \nabla p/\rho + \nu \nabla^2 \mathcal{V}]$.
3. Marker particles are moved with the new velocity to track the free surface.

3.4 Fractional-step (Chorin projection) method

Step 1 Intermediate velocity ignoring pressure: $\mathcal{V}^* = \mathcal{V}^n + \Delta t[-(\mathcal{V}^n \cdot \nabla)\mathcal{V}^n + \nu \nabla^2 \mathcal{V}^n]$

Step 2 Pressure Poisson: $\nabla^2 p^{n+1} = (\rho/\Delta t) \nabla \cdot \mathcal{V}^*$

Step 3 Projection (correction): $\mathcal{V}^{n+1} = \mathcal{V}^* - (\Delta t/\rho) \nabla p^{n+1}$ -> $\nabla \cdot \mathcal{V}^{n+1} = 0$

The pressure projects $\mathcal{V}^*$ onto the space of divergence-free fields. Time step limits (explicit): $\Delta t \le \Delta x^2/4\nu$ and $\Delta t \le 4\nu/u^2$ (also CFL).

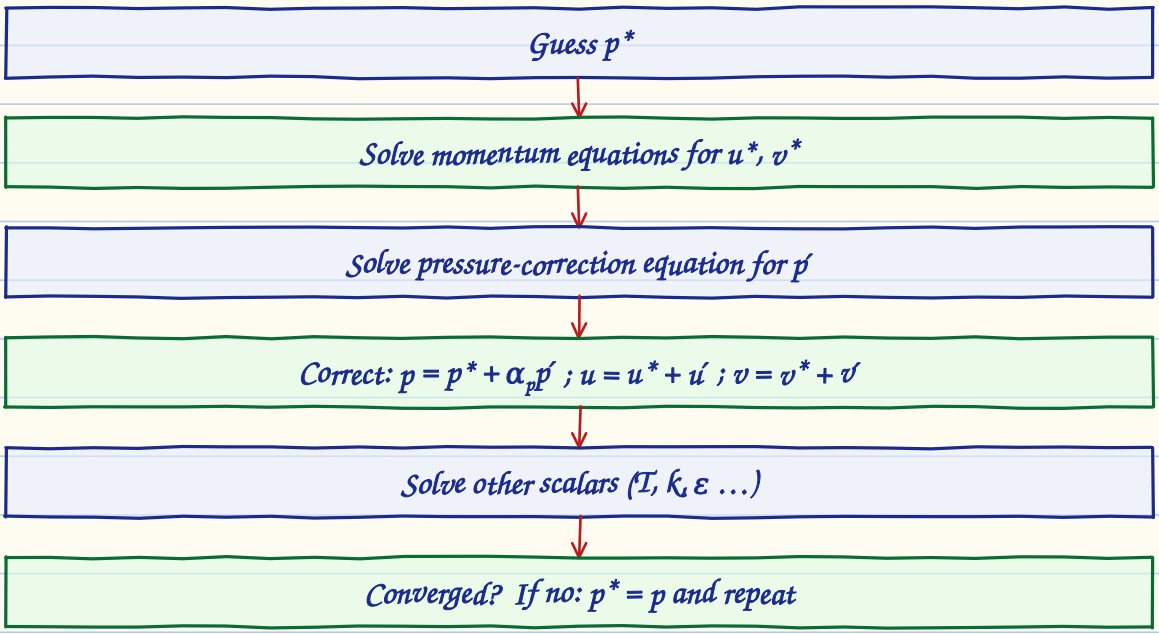
4. Pressure Correction Methods — SIMPLE family

Used in finite-volume/finite-difference codes for steady (and unsteady) flow. Idea: guess pressure, solve momentum, then correct pressure and velocity so that continuity is satisfied.

SIMPLE = Semi-Implicit Method for Pressure-Linked Equations (Patankar & Spalding, 1972)

$$p = p^* + p' , \quad u = u^* + u' , \quad v = v^* + v'$$

- Starred values: from the momentum equations with guessed pressure p^*. Primed values: corrections.
- Velocity correction: $u' = -d_e (p'_E - p'_P)$, $d_e = A_e/a_e$ (neglecting neighbour corrections — the "simplification" in SIMPLE).
- Substitute into continuity -> pressure-correction equation: $a_P p'_P = a_E p'_E + a_W p'_W + a_N p'_N + a_S p'_S + b$, b = mass imbalance of u^*, v^*.



- Under-relaxation is essential: $\alpha_p \approx 0.3$ for pressure and $\alpha_u \approx 0.7$ for velocity (steady problems); $\alpha_p + \alpha_u \approx 1$ for optimum.
- SIMPLER: solves a pressure equation (not correction) for p, uses p′ only to correct velocities; better convergence.
- SIMPLEC: approximate d by $\mathcal{A}/(a_P - \Sigma a_{nb})$ — keeps neighbour corrections; α_p can be 1; faster.
- PISO (Pressure-Implicit with Splitting of Operators): one predictor + two or more corrector steps per time step; best for transient flows, no under-relaxation needed.

Method	Correctors	Under-relax	Best use
SIMPLE	1	needed	steady flow
SIMPLER	1 + p-eq	needed	steady, better conv.
SIMPLEC	1 (consistent)	light	steady
PISO	2 or more	not needed	transient

Discretised momentum equation in SIMPLE (finite-volume form)

$$a_e u_e = \Sigma\, a_{nb} u_{nb} + b + (p_P - p_E)\mathcal{A}_e$$

a_{nb} from convection–diffusion (upwind / hybrid / power law); $(p_P - p_E)\mathcal{A}_e$ = pressure force across the east face of the u control volume (u-cell is staggered). Pressure-correction coefficients:

$$a_E = \rho_e d_e \mathcal{A}_e, \quad a_W = \rho_w d_w \mathcal{A}_w, \quad a_P = a_E + a_W + a_N + a_S, \quad b = (\rho u^*)_w \mathcal{A}_w - (\rho u^*)_e \mathcal{A}_e + (\rho v^*)_s \mathcal{A}_s - (\rho v^*)_n \mathcal{A}_n$$

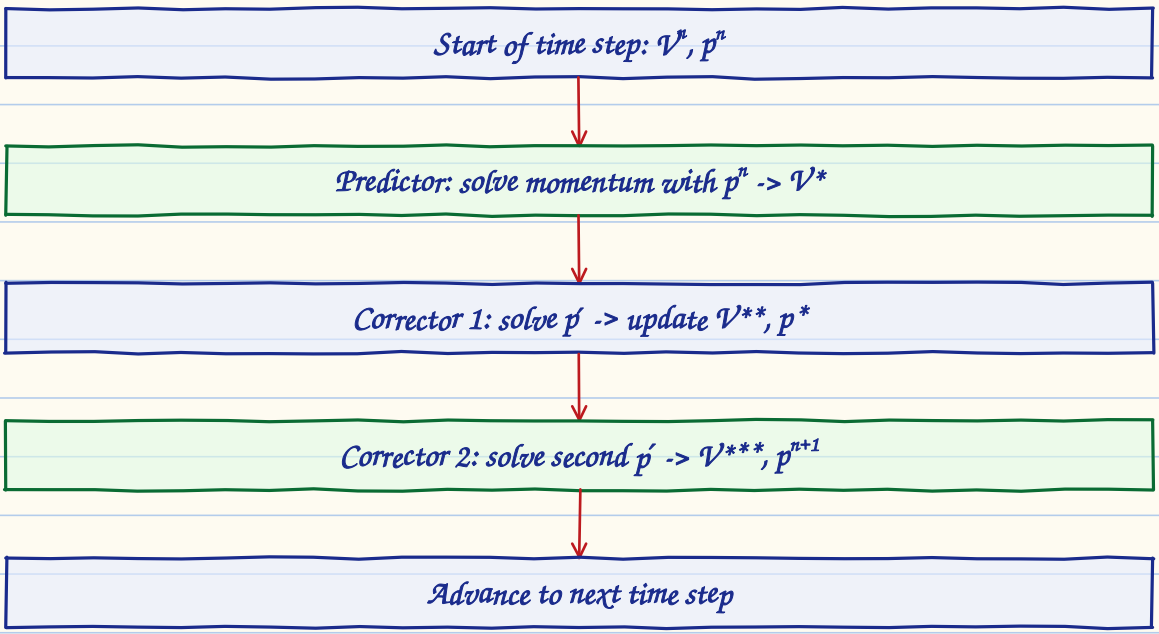
- b is the mass source: when $b \to 0$ everywhere, continuity is satisfied and the solution is converged.
- Convergence monitors: residuals of u, v, mass imbalance ($\Sigma |b|$), and key quantity such as drag. Typical criterion: normalised residual < 10^{-4} to 10^{-6}.

- Boundary conditions for p' : at an inlet/wall where velocity is specified, $u' = 0$ so no p' equation is needed (Neumann).

Example: 1-D pressure correction

Mass imbalance gives p'

Cell with $u^*_w = 1.0$, $u^*_e = 1.2$ ($\rho = 1$, $A = 1$). $b = 1.0 - 1.2 = -0.2$ (more leaves than enters). With $d_e = d_w = 0.5$: $a_E = a_W = 0.5$, $a_P = 1.0$ and $p'_E = p'_W = 0$: $p'_P = -0.2$. Negative p'_P reduces the outflow and increases inflow -> mass imbalance is removed.



5. Vortex Methods

Lagrangian, grid-free approach: the flow is represented by a collection of discrete vortices (vortex blobs) that move with the local velocity. Pressure does not appear.

- Vorticity transport equation (2-D): $D\omega/Dt = \nu\nabla^2\omega$. Inviscid part: vorticity is carried by the flow; viscous part: diffusion of vorticity.
- Discretise: $\omega(x) \approx \Sigma_j \Gamma_j \zeta_\sigma(x - x_j)$ with circulation Γ_j and smoothing core ζ_σ.
- Velocity from the Biot–Savart law:

$$V(x) = \Sigma_{j=1}^{N} \Gamma_j K_\sigma(x - x_j), \quad K(x) = (1/2\pi)(-y, x)/|x|^2$$

- Regularised kernel (Chorin blob): $K_\sigma = K\cdot(|x|^2/(|x|^2 + \sigma^2))$ to avoid singularity when vortices come close.
- Convection step: $dx_i/dt = V(x_i)$, integrated by RK2 / Euler. Cost $O(N^2)$ — reduced by fast multipole / tree codes.

Viscous effects in vortex methods

- **Random-walk method** (Chorin): each blob is given a random displacement $\Delta x, \sim \mathcal{N}(0, 2\nu\Delta t)$ each step — stochastic simulation of diffusion.
- Core-spreading and particle-strength-exchange (PSE) methods give deterministic diffusion.
- Boundary layers: new vortex sheets are created at a wall to satisfy no-slip.
- Vortex-in-cell (VIC): vorticity assigned to a grid, Poisson equation $\nabla^2\psi = -\omega$ solved on the grid (FFT) and velocities interpolated back — $O(N \log N)$.

Features

- Advantages: no numerical diffusion, no grid for the flow region (good for external flows, wakes, high Re), automatic far-field BC.
- Limitations: expensive N-body sum, boundary treatment, statistical noise in random walk, compressibility not included.

Algorithm of a vortex-blob code

Initialise blobs (positions x_j, circulations Γ_j)

↓

Compute velocity at each blob by Biot–Savart sum

↓

Move blobs: $x_j \leftarrow x_j + V\Delta t$ (RK2)

↓

Add random walk $\xi \sim \mathcal{N}(0, 2\nu\Delta t)$ for viscosity

↓

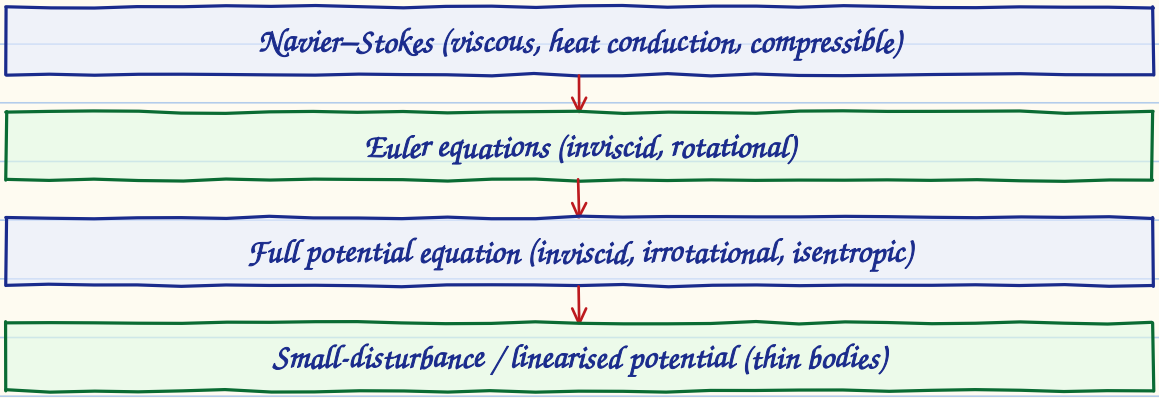
Create new vortices at the wall (no-slip) and merge/remove blobs

Example: two equal point vortices

Two identical vortices of circulation Γ at distance d. Each induces speed $V = \Gamma/(2\pi d)$ on the other, perpendicular to the line joining them.

They rotate about their midpoint with angular velocity $\Omega = V/(d/2) = \Gamma/(\pi d^2)$. Opposite signs (vortex pair) translate together with speed $\Gamma/(2\pi d)$.

6. Compressible Flows — Hierarchy of Models



6.1 Potential equation

For irrotational flow $\mathbf{V} = \nabla\varphi$, isentropic, steady, the continuity equation becomes the full potential equation:

$$(a^2 - \varphi_x^2)\varphi_{xx} + (a^2 - \varphi_y^2)\varphi_{yy} - 2\varphi_x\varphi_y\varphi_{xy} = 0\,, \quad a^2 = a_0^2 - ((\gamma-1)/2)(\varphi_x^2 + \varphi_y^2)$$

- Single non-linear second-order PDE. Type: elliptic for subsonic ($M < 1$), hyperbolic for supersonic ($M > 1$) — a mixed type problem for transonic flow.
- Linearised small-disturbance form $(1 - M_\infty^2)\varphi_{xx} + \varphi_{yy} = 0$ (Prandtl–Glauert).
- Type-dependent differencing (Murman–Cole): central differences in subsonic points; upwind (retarded density) in supersonic points; solved by line SOR; conservative form needed to capture shocks. Isentropic assumption overestimates shock strength.

6.2 Euler equations

Inviscid, conservative form (2-D):

$$\partial U/\partial t + \partial F/\partial x + \partial G/\partial y = 0$$

$$U = (\rho, \rho u, \rho v, E)^T\,, \quad F = (\rho u, \rho u^2 + p, \rho uv, u(E + p))^T\,, \quad G = (\rho v, \rho uv, \rho v^2 + p, v(E + p))^T$$

$$E = \rho[e + (u^2 + v^2)/2]\,, \quad p = (\gamma - 1)[E - \tfrac{1}{2}\rho(u^2 + v^2)]$$

- Hyperbolic in time with eigenvalues of the Jacobian $A = \partial F/\partial U$: $u,\ u,\ u + a,\ u - a$ ($a = \sqrt{(\gamma p/\rho)}$).
- Solved by explicit MacCormack, Lax–Wendroff, Jameson RK schemes with artificial dissipation, or upwind / Godunov / Roe schemes (see Unit II, IV). CFL: $\Delta t \le CFL\cdot\Delta x/(|u| + a)$.

6.3 Navier–Stokes system (compressible)

$$\partial U/\partial t + \partial F/\partial x + \partial G/\partial y = \partial F_v/\partial x + \partial G_v/\partial y$$

$$F_v = (0, \tau_{xx}, \tau_{xy}, u\tau_{xx} + v\tau_{xy} - q_x)^T\,, \quad G_v = (0, \tau_{xy}, \tau_{yy}, u\tau_{xy} + v\tau_{yy} - q_y)^T$$

$$\tau_{xx} = \mu(4u_x - 2v_y)/3\,, \quad \tau_{yy} = \mu(4v_y - 2u_x)/3\,, \quad \tau_{xy} = \mu(u_y + v_x)\,, \quad q = -k\nabla T$$

- Closed with perfect-gas law and Sutherland viscosity $\mu(T)$. Convective part treated as hyperbolic

(MacCormack / upwind), viscous part by central differences.

- Explicit MacCormack for N–S: predictor-corrector with forward/backward differences on convective terms, alternating on viscous derivatives (cross derivatives central). Time step: $\Delta t \le \min(\Delta t_{CFL}, \Delta t_{visc})$, $\Delta t_{visc} \sim \Delta x^2 \rho / 2\mu$.

7. Flow-Field-Dependent Variation (FDV) Methods

Developed by T. J. Chung. Aim: a single finite element / finite volume algorithm valid for the whole range of flows: incompressible → compressible, laminar → turbulent, subsonic → hypersonic.

1. Expand the unknown in a Taylor series in time including second-order term: $U^{n+1} = U^n + \Delta t\, U_t^n + (\Delta t^2/2)\, U_{tt}^n + \dots$
2. Replace the time derivatives by spatial derivatives using the governing equations ($U_t = -\nabla \cdot \mathcal{F} + \dots$, $U_{tt} = \dots$).
3. Introduce flow-field-dependent variation parameters s_1, s_2 (convection) and s_3 (diffusion) that are computed from the local gradients of the flow variables between two iterations: $\Delta U/\Delta t$ and $\Delta \mathcal{F}/\Delta U$ — i.e. from the change of the flow field itself.
4. The resulting terms act as automatic, solution-dependent artificial viscosity — large at shocks and steep gradients, small in smooth regions — so no ad-hoc dissipation coefficients are required.
5. Equations are solved on a finite element (or finite volume) mesh with the Galerkin method and an implicit/explicit time integration (Newton / GMRES).

- Result: shock capturing without user-tuned smoothing, a unified treatment of pressure–velocity coupling (compressibility enters through the equation of state), efficient on unstructured meshes.

Shock relations (normal shock) — useful for checks

$$p_2/p_1 = 1 + (2\gamma/(\gamma+1))(M_1^2 - 1), \quad \rho_2/\rho_1 = (\gamma+1)M_1^2/((\gamma-1)M_1^2 + 2)$$

$$M_2^2 = (1 + (\gamma-1)M_1^2/2) / (\gamma M_1^2 - (\gamma-1)/2)$$

Numerical solutions are validated against these Rankine–Hugoniot values; conservative schemes reproduce them, non-conservative schemes do not.

Artificial viscosity for central/Lax–Wendroff-type Euler solvers

- Second-difference term (shock capturing): $\varepsilon^{(2)}(U_{i+1} - 2U_i + U_{i-1})$, $\varepsilon^{(2)} \propto |p_{i+1} - 2p_i + p_{i-1}|/(p_{i+1} + 2p_i + p_{i-1})$ — large only across shocks.
- Fourth-difference term (background damping in smooth regions): $-\varepsilon^{(4)}(U_{i+2} - 4U_{i+1} + 6U_i - 4U_{i-1} + U_{i-2})$. Jameson–Schmidt–Turkel (JST) scheme uses both with RK time stepping.

MacCormack algorithm for 2-D Euler / N–S

Predictor: $\bar{U} = U^n - (\Delta t/\Delta x)\Delta F^n_{forward} - (\Delta t/\Delta y)\Delta G^n_{forward}$

↓

Evaluate $F, \bar{G}$ from $\bar{U}$ (new fluxes)

↓

Corrector: $U^{n+1} = \frac{1}{2}[U^n + \bar{U} - (\Delta t/\Delta x)\Delta F_{backward} - (\Delta t/\Delta y)\Delta \bar{G}_{backward}]$

↓

Add artificial viscosity; apply BCs; set Δt from CFL

Short answer questions

Q1. Why is the pressure difficult in incompressible flow? -> no evolution equation; it only enforces $\nabla \cdot V = 0$.

Q2. Why a staggered grid? -> avoids checkerboard pressure fields by coupling p and u directly.

Q3. Why under-relaxation in SIMPLE? -> p' overcorrects (neglected neighbour terms) leading to divergence.

Q4. Advantage of vortex methods? -> grid-free, no numerical diffusion, pressure eliminated.

Q5. Type of the potential equation? -> elliptic if $M < 1$, hyperbolic if $M > 1$ (mixed in transonic flow).

Q6. Why conservation form for Euler equations? -> correct shock speeds/strengths (Rankine–Hugoniot).

8. Boundary Conditions for Flow Problems

Boundary	Condition	Remarks
Solid wall (viscous)	$u = v = 0$; $T = T_w$ or $\partial T/\partial n = 0$	no-slip; adiabatic / isothermal
Solid wall (inviscid)	$V \cdot n = 0$	slip; tangency
Symmetry plane	$V \cdot n = 0$, $\partial \varphi/\partial n = 0$	half-domain only
Inflow (subsonic)	specify ρ, u, v (or total p, T)	one variable from inside
Outflow (subsonic)	specify static p	extrapolate rest
Supersonic inflow	all variables fixed	characteristics all enter
Supersonic outflow	extrapolate all	no info from outside
Far-field	characteristic / freestream	non-reflecting

- Number of conditions = number of characteristics entering the domain (from the sign of eigenvalues u, $u \pm a$).
- Incompressible pressure: Neumann at walls $\partial p/\partial n = \ldots$; at least one Dirichlet value (or reference) needed since p is defined up to a constant.
- Numerical treatment: ghost cells, one-sided differences, extrapolation $\varphi_N = 2\varphi_{N-1} - \varphi_{N-2}$; wall pressure from normal momentum $\partial p/\partial n = 0$ (boundary layer).

9. Example Problems

Example 1 — Lid-driven cavity (incompressible, ψ–ω)

Square cavity, lid speed $U = 1$ along $y = 1$, $Re = UL/\nu = 100$. Boundary: $\psi = 0$ on all walls; ω on walls from Thom's formula (lid: $-2\psi_1/\Delta y^2 - 2U/\Delta y$).

Grid 41×41, $\Delta t = 0.005$ (explicit, $\Delta t \le \Delta x^2/4\nu$), SOR with $\omega_s \approx 1.8$ for Poisson. Result: primary vortex centre near (0.62, 0.74), small corner vortices at the bottom corners (Ghia et al. benchmark).

Example 2 — Shock tube (Sod problem), 1-D Euler

Initial: left ($x < 0.5$): $\rho = 1$, $u = 0$, $p = 1$; right: $\rho = 0.125$, $u = 0$, $p = 0.1$; $\gamma = 1.4$.

Solution contains a left-moving rarefaction fan, a contact discontinuity and a right-moving shock. MacCormack with artificial viscosity or TVD scheme captures all waves; Δt from CFL ≤ 0.8.

Example 3 — Supersonic flow past a wedge

$M_\infty = 2$, wedge half-angle 10°: oblique shock (θ-β-M relation) gives $\beta \approx 39.3°$, downstream $M \approx 1.64$. Euler solver marching in x (hyperbolic in supersonic flow); wall: $V \cdot n = 0$; inflow all fixed; outflow extrapolated.

Quick revision — Unit III

- Incompressible: ψ–ω (no pressure, wall ω from Thom), MAC/projection (Poisson for p), SIMPLE ($p = p^* + p'$, under-relaxation).
- Vortex methods: Biot–Savart law, blobs, random walk for viscosity, grid-free.
- Compressible: potential (elliptic/hyperbolic mixed) → Euler (hyperbolic) → Navier–Stokes; eigenvalues u, $u \pm a$.
- FDV: Taylor-series in time with flow-dependent parameters s_1, s_2, s_3 = built-in shock-capturing dissipation.
- BCs: number of conditions = number of characteristics entering.

Important Questions

1. Derive the vorticity–stream function formulation and explain how wall vorticity is obtained.
2. Explain the SIMPLE algorithm with a flow chart; compare with SIMPLER, SIMPLEC and PISO.
3. What is the staggered grid and why is it used? Explain the projection method.
4. Explain vortex methods with Biot–Savart law and the treatment of viscosity.
5. Write the Euler and Navier–Stokes equations in conservation form; explain FDV methods and boundary conditions.